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REG NO :CIT-221-086/2024
UNIT:DATABASE
FACULTY:COMMUNICATION INFORMATION AND TECHNOLOGY

I)CREATE TABLE STUDENT

```
CREATE TABLE Student (
   RegNo VARCHAR(20) PRIMARY KEY,
   SurName VARCHAR(50),
   FirstName VARCHAR(50),
   Gender VARCHAR(10),
   YearStudy INT,
   HostelID VARCHAR(10),
   FOREIGN KEY (HostelID) REFERENCES Hostel(HostelID)
);
```

ii) Add two records into Student

```
INSERT INTO Student
VALUES ('DIBM101', 'Malt', 'Guroe', 'Male', 3, 'D106');

INSERT INTO Student
VALUES ('DICE204', 'Miriam', 'Risper', 'Female', 1, 'B202');
```

iii) Display students in Hostel D106

```
SELECT *
FROM Student
WHERE HostelID = 'D106';
```

2) University Database Case Study

i) ER Diagram (Description)

Entities:

- **Student** (StudentID, Name, Email, Major)
- **Course** (CourseID, Title, Credits)
- **Enrollment** (Grade)

Relationship:

- Student **enrolls in** Course
- Many-to-Many (M:N)
- Enrollment is associative entity

ER Structure (Text Diagram)

```
Student (1) --------< Enrollment >-------- (1) Course
        (M)                (M)
```

ii) Relational Schema

Student

- StudentID (PK)
- Name
- Email
- Major

Course

- CourseID (PK)
- Title
- Credits

Enrollment

- StudentID (PK, FK)
- CourseID (PK, FK)
- Grade

iii) Students enrolled in “Database Systems”

```
SELECT S.Name
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
JOIN Course C ON C.CourseID = E.CourseID
WHERE C.Title = 'Database Systems';
```

iv) Courses with more than 5 students

```
SELECT C.Title, COUNT(*) AS TotalStudents
FROM Course C
JOIN Enrollment E ON C.CourseID = E.CourseID
GROUP BY C.Title
HAVING COUNT(*) > 5;
```

v) Average grade per course

```
SELECT C.Title, AVG(E.Grade) AS AverageGrade
FROM Course C
JOIN Enrollment E ON C.CourseID = E.CourseID
GROUP BY C.Title;
```

vi) Normalization to 3NF

The database is already in **3NF** because:

1. Each table has a primary key.
2. No partial dependencies (Enrollment uses composite PK properly).
3. No transitive dependencies (non-key attributes depend only on PK).

Therefore:

- Student → attributes depend only on StudentID.
- Course → attributes depend only on CourseID.
- Enrollment → Grade depends on composite key.

vii) Referential Integrity Implementation

Use **Foreign Key constraints**:

```
FOREIGN KEY (StudentID) REFERENCES Student(StudentID)
ON DELETE CASCADE
ON UPDATE CASCADE;
FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
ON DELETE CASCADE
ON UPDATE CASCADE;
```

This ensures:

- Cannot enroll non-existing student.
- Cannot enroll in non-existing course.
- Prevents orphan records.

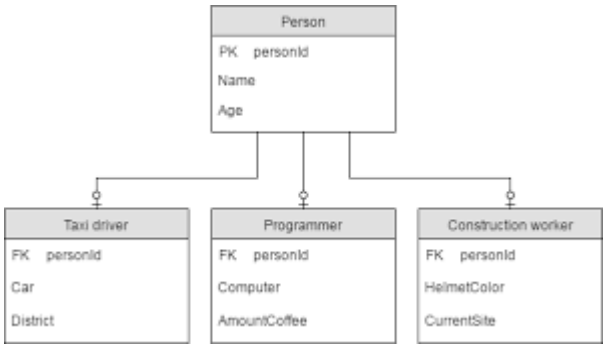
3) Four Properties of a Transaction (ACID)

1. **Atomicity** – All operations complete or none do.
2. **Consistency** – Database remains valid after transaction.
3. **Isolation** – Transactions do not interfere.
4. **Durability** – Once committed, data is permanent.

4) Three Recovery Facilities

1. **Transaction Log (Write-Ahead Logging)**
   Records all changes before execution.
2. **Checkpointing**
   Saves database state periodically to reduce recovery time.
3. **Backup and Restore**
   Full and incremental backups used after failure.

5) Hierarchical Database Model



4

- Data organized as a **tree structure**
- One parent → many children
- Each child has only one parent
- Example: IBM IMS system

Structure:

Parent
├— Child 1
├— Child 2

6) Differences

i) Data Warehouse vs Data Mining

Data Warehouse	Data Mining
Stores large historical data	Extracts patterns from data
For reporting & analysis	For prediction & insights

ii) Super Key vs Candidate Key

- **Super Key** – Any attribute(s) that uniquely identify a record.
- **Candidate Key** – Minimal super key (no redundant attributes).

iii) Entity vs Attribute

- **Entity** – Object in real world (e.g., Student).
- **Attribute** – Property of entity (e.g., Name).

7) Company ER Diagram

Entities:

- Department
- Employee
- Manager (is-a Employee)
- Payroll
- SportsClub

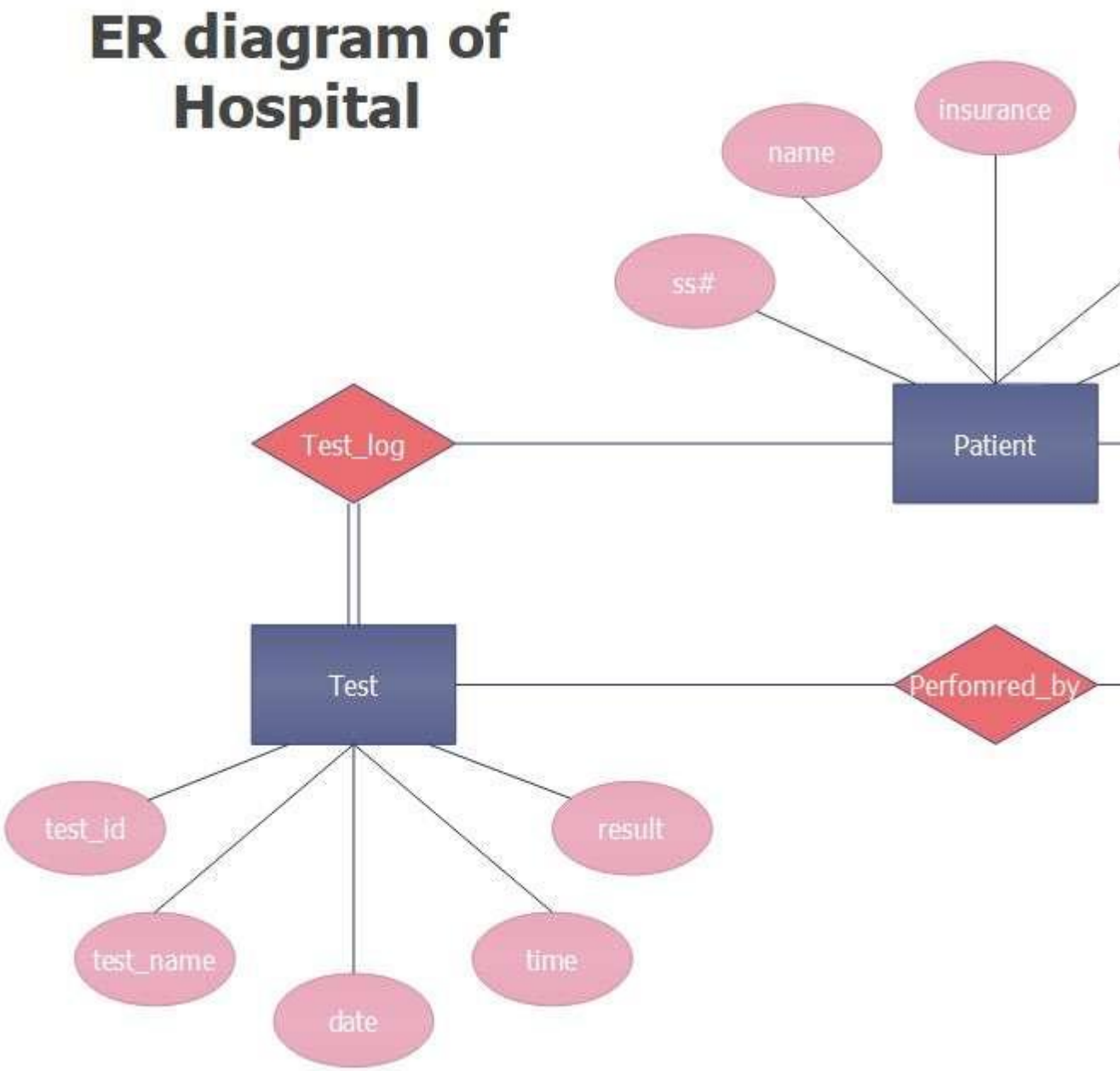
Relationships:

- Department employs Employees (1:M)
- Department has Manager (1:1)
- Employee paid via Payroll (Monthly/Weekly)
- Employee may join SportsClub (optional)

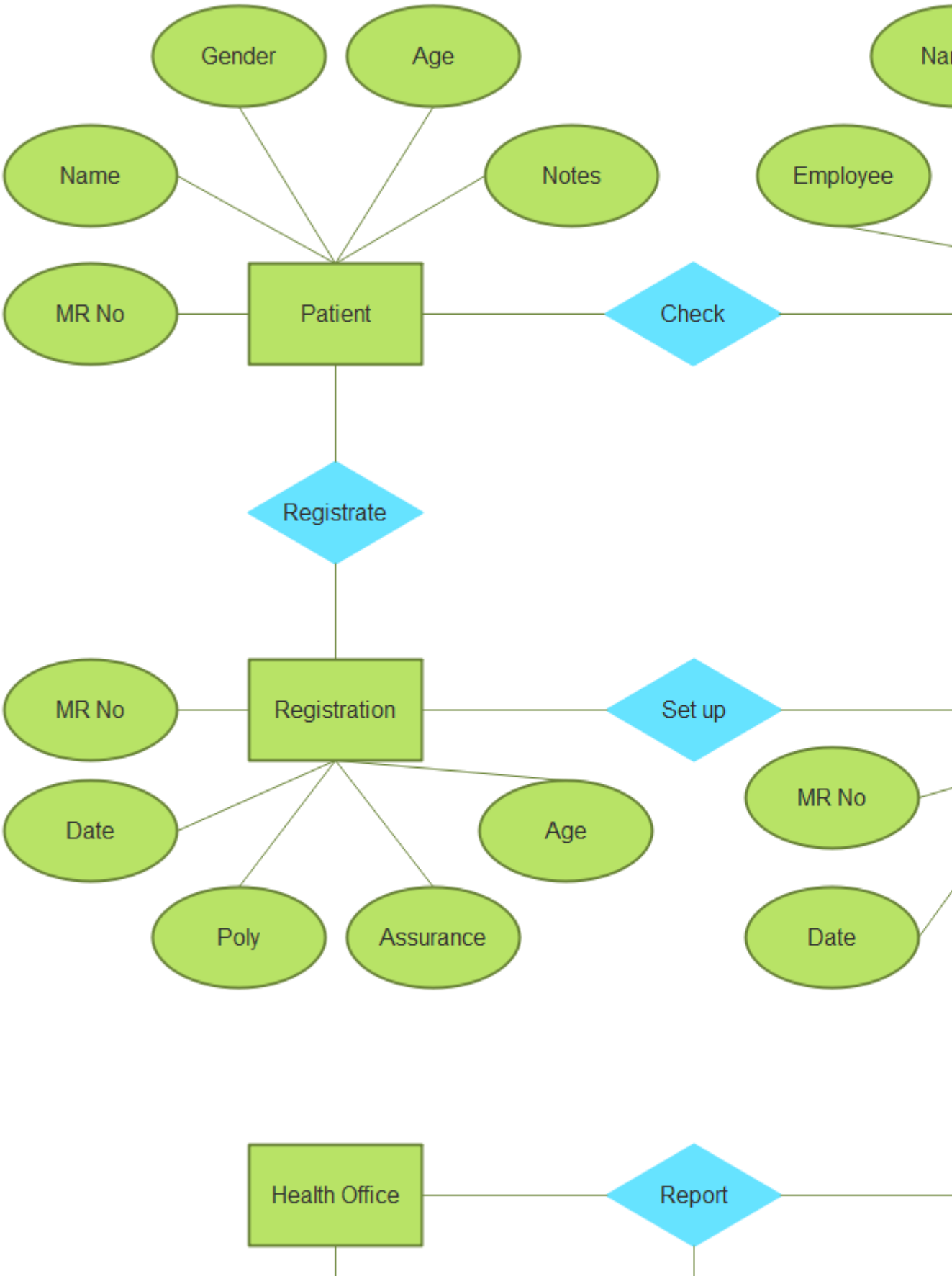
Text Representation:

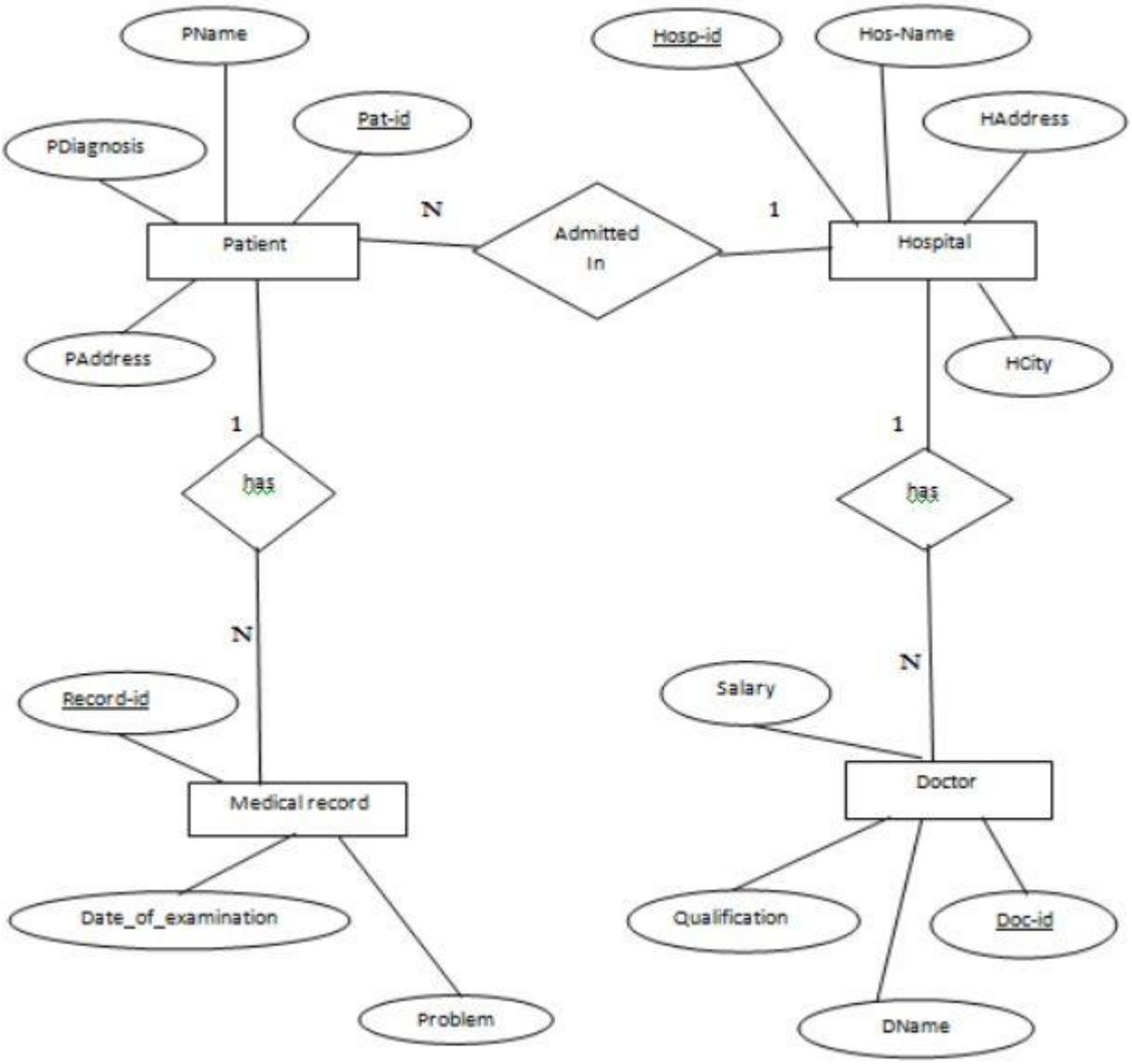
Department (1) ------ (M) Employee
Department (1) ------ (1) Manager
Employee (1) ------ (1) Payroll
Employee (0..1) ---- (1) SportsClub

8) Hospital ER Diagram



ER Diagram of Hospital Information





Entities:

- Patient (In/Out subtype)
- Doctor
- Nurse
- Ward
- Bed
- Theatre

Relationships:

- Doctor treats Patient (M:N)
- Nurse assigned to Ward (1:M)
- Ward contains Beds (1:M)

- Patient admitted to Ward (M:1)
- Theatre used for operations

9) Client/Server vs Centralized Database

Client/Server	Centralized
Processing shared	Processing at one location
More scalable	Simple management

Advantage:

- Client/Server → Better performance.
- Centralized → Easier security control.

10) ACID and Concurrency/Recovery

- **Atomicity & Durability** → Achieved using logs & recovery systems.
- **Isolation** → Managed by locking & concurrency control.
- **Consistency** → Enforced through constraints and integrity rules.

11) Three Database Anomalies

1. **Insertion Anomaly** – Cannot insert data without other data.
2. **Deletion Anomaly** – Deleting data removes unintended data.
3. **Update Anomaly** – Same data must be updated in multiple places.

Normalization removes redundancy and eliminates these anomalies.